USER GUIDE

Demand Functions

Where the demand curve comes from. Move a price, the optimal bundle moves, and the trail it leaves *is* the demand function.

BEFORE YOU START

Opening it

There is nothing to install. It is a web page: it opens in Chrome, Safari, Edge or Firefox, on a computer, tablet or phone. You need a connection the first time; after that the browser keeps it and it works offline.

Top right there are two pairs of buttons. The first switches the language between **ES** and **EN**; the second goes from **Dark** to **Light**. For projecting in class, light mode reads better on a beamer; on screen, dark is easier on the eyes.

> **Every graph zooms.** Mouse wheel over the graph, or a two-finger pinch on the trackpad or touchscreen. The zoom applies to the graph under the pointer only, not to the whole page.

> Each slider has a number box beside it. For an exact value, type it into the box and press `Enter`; the slider is for exploring, the box is for precision.

WHAT IS ON SCREEN

The panels

On the left, the choice: the (x, y) plane with the budget line and the indifference curve through the optimum. On the right, the demand: the same optimum, drawn against whichever variable you are moving.

PANEL	WHAT IT SHOWS
Choice · (x, y) plane	Budget line, indifference curve and the optimal point. Everything else comes from here.
Demand	The optimum drawn against p_x, p_y or income. If you move income the panel is titled *Engel curve*, which is its proper name.

> You can show one panel or both. One alone is bigger, and for projecting in class that is usually better.

The readouts

READOUT	HOW TO READ IT
x*, y*	The optimal bundle at the current prices and income.
Spend	$p_x \cdot x^* + p_y \cdot y^*$. With monotone preferences it equals income.
Type	Whether x is normal, inferior or Giffen **at the point you are at**. It is not a property of the whole function: it changes from stretch to stretch.

RANGES: THE FIRST THING TO SET

Why those boxes are there

At the very top there are three pairs of boxes: p_x from … to …, p_y from … to …, m from … to … Each pair does two things at once: it sets how far the slider travels, and it sets the stretch the horizontal axis of the demand panel draws.

> A minimum price near zero ruins the picture. Demand for x runs off to infinity as $p_x \to 0$, so the vertical axis stretches until the interesting part of the curve is squashed flat against the axis. Leave the minimum at 1, or at least 0.5.

The fixed vertical axis

With **Fixed vertical axis** on, the ceiling of the vertical axis is m_{max} divided by the minimum price, and it never moves. That is the most anyone could buy given the ranges you set, so the curve always fits and always reaches down to zero. Turn it off and the axis fits the current curve: bigger, but it jumps every time you touch a parameter and comparisons stop being fair.

> To compare two situations — before and after a price rise, or two preference families — the fixed axis is essential. If the axis moves, the two curves are drawn at different scales and the comparison lies.

THE PREFERENCE FAMILIES

Six starting points

FAMILY	WHAT IT IS AND ITS PARAMETERS
Cobb–Douglas	$U = x^{\alpha} \cdot y^{1-\alpha}$. One parameter, α. Demand is $x^* = \alpha \cdot m/p_x$: a hyperbola, and spending on x does not depend on the price.
CES	$U = \gamma_H \cdot (\gamma \cdot x^{\rho} + (1-\gamma) \cdot y^{\rho})^{1/\rho}$, where γ_H is the harmonic mean of γ and 1−γ. Two parameters: γ splits the weight, ρ governs substitutability. At ρ = 0 it becomes Cobb–Douglas.
Quasilinear	First you choose which variable is the linear one — **u = φ(x) + y** or **u = x + ψ(y)** — and then you type the curved half into the box that appears.
Linear (substitutes)	U = a·x + y. MRS is constant at a. Demand is all-or-nothing: a jump from one axis to the other when p_x/p_y crosses a.
X inferior (Giffen)	$U = \ln(x - \underline{x}) - ((1+b)/b) \cdot \ln(\bar{y} - y)$, with $b = (1-\beta)/\beta$. Three parameters: β, the subsistence floor $\underline{x}$ and the ceiling $\bar{y}$. This is the family you hunt the Giffen case with.
Custom	Type your own utility. The optimum is found numerically, so kinks and corners are fine.

> The quasilinear one has two steps on purpose. Until you say which variable is linear, the function box does not appear: φ may only depend on x, and ψ only on y. If you type `ln(y)` into the φ box, the applet refuses it.

DISCOVERING A DEMAND CURVE

The idea

The demand curve is not postulated: it is built. Hold everything fixed but one variable, move that variable, and record the optimal bundle each time. The **Discover the demand curve** group does exactly that, and lets the student watch it happen rather than being told.

CONTROL	WHAT IT DOES
Variable being swept	Which one you move: p_x, p_y or m. The other two stay fixed. Recorded points belong to the chosen variable only: change it and you start a fresh collection.
Mark while moving	On, every touch of the slider leaves a mark. This is the quick way.
Record point	Marks only when you press it. This is the slow way, and the better one for explaining: move, ask, record.
Sweep / Stop	The applet sweeps the variable from end to end on its own, leaving marks. For the finish, once it is clear where the points come from.
Clear	Clears every recorded point and starts over.

A ten-minute run

1. Pick **Cobb–Douglas**. Set the ranges: p_x from **0.5** to **6**, p_y from **0.5** to **4**, m from **0** to **10**. Type the numbers into the boxes and press `Enter`.

2. Set α = **0.5**, p_y = **2**, m = **6**. Leave p_x at 1.

   ```
   x* = 3, y* = 1.5. Half the income on each good.
   ```

3. Turn off **Mark while moving** and **Reveal the curve**. The swept variable should be p_x. Raise p_x to 1.5 and press **Record point**. Repeat at 2 and at 3.

   Four points: (1, 3), (1.5, 2), (2, 1.5), (3, 1). That is exactly $x^* = 3/p_x$.

4. Switch on **Join the points**. A polyline appears that already looks like a hyperbola. Now switch on **Reveal the curve** to see the exact curve underneath.

   ```
   The points land on the curve. No trick: it is the same computation.
   ```

5. Watch spending on x as you move p_x: $p_x \cdot x^* = 3$ always. That is the Cobb–Douglas signature, and it explains why the curve is a hyperbola.

6. Change the swept variable to **m** and clear the points. Sweep income from 0 to 10.

   The Engel curve is a straight line through the origin: $x^* = 0.5 \cdot m/p_x$.

What it takes

A Giffen good is one whose demand *rises* when its own price rises. It is rare, and it needs two things at once: the good must be inferior, and it must weigh so heavily in the budget that the income effect swallows the substitution effect. Being inferior is not enough.

> In the **X inferior** family the Giffen stretch appears once income exceeds $p_y \cdot \bar{y}$. With that inequality the other way round, x is inferior but not Giffen — which is exactly the distinction worth teaching.

A recipe that works

1. Pick the **X inferior (Giffen)** family.

2. Ranges: p_x from **7** to **21**, p_y from **0.5** to **4**, m from **0** to **10**.

3. Parameters: β = **0.89**, $\underline{x}$ = **0.3**, $\bar{y}$ = **5**. Prices and income: p_y = **1**, m = **7**.

   The condition holds: $m = 7 > p_y \cdot \bar{y} = 5$.

4. Switch on **Highlight Giffen stretch** and **Reveal the curve**, and sweep p_x from end to end.

   Demand for x rises: 0.302 at $p_x = 7$, 0.312 at 10, 0.321 at 15 and 0.325 at 21. The curve slopes upward and the stretch is highlighted.

5. Now drop income to **4**, below $p_y \cdot \bar{y} = 5$, and sweep again.

   The highlight disappears: x is still inferior, but no longer Giffen.

> If the demand panel goes blank over part of the sweep, you are outside the domain of the function: this family needs $x > \underline{x}$ and $y < \bar{y}$. Raise the minimum price or lower income until the curve is whole again.

THE LAYERS

What each one turns on

LAYER	WHAT IT DRAWS
Recorded points	The marks you have left.
Join the points	A polyline from mark to mark.
Reveal the curve	The exact demand curve. Leave it off while the student builds it.
Price-consumption path	In the (x, y) plane, the trail of the optimum as the price moves. The same information as the demand curve, in other coordinates.
Budget line	The constraint in the (x, y) plane.
Indifference curve	The one through the current optimum.
Demand for x	The curve for x in the demand panel.
Demand for y	The one for y, in the same panel. It is usually the surprising one.
Highlight Giffen stretch	Colours the stretch where demand rises with the price.
Grid	The background grid.

> A nice example with the quasilinear: choose **u = φ(x) + y**, type `4ln(x)`, set $p_y = 2$ and $m = 10$. Demand for x is $x^* = 4p_y/p_x = 8/p_x$, and demand for y comes out **flat**: whatever p_x is, $y^* = 1$. Spending on x is always $4p_y = 8$.

WRITING YOUR OWN FUNCTION

The syntax it accepts

There are two boxes: the custom utility one (variables **x** and **y**) and the curved half of the quasilinear one (only **x** if you chose φ, only **y** if you chose ψ).

CONTROL	WHAT IT DOES
`+ − * / ^`	Add, subtract, multiply, divide, power. `x^0.5` is the square root of x.
`( )`	Brackets. `(x+y)^2` is not the same as `x+y^2`.
`sqrt ln log exp abs`	Root, natural log, base-10 log, exponential and absolute value.
`min max`	Two arguments: `min(x,y)` is perfect complements.
`sin cos`	Trigonometric, in radians. Rarely useful here, but available.
`e pi`	The two constants. `e` is 2.718…, `pi` is 3.1416…
`2x`, `3xy`	Implicit multiplication works: `2x` reads as `2*x`.

Examples that work

EXPRESSION	WHAT PREFERENCES IT DESCRIBES
`x^0.5*y^0.5`	Symmetric Cobb–Douglas, as a check.
`4ln(x)`	In φ(x): demand comes out $x^* = 4p_y/p_x$, independent of income.
`2sqrt(x)`	In φ(x): $x^* = p_y^2/p_x^2$.
`3ln(y)`	In ψ(y): now x is the linear good, and the flat demand is x's.
`min(x,y)`	In the custom box: perfect complements, demand $m/(p_x+p_y)$.

If it cannot be read you get *"I can't read that expression"* and the graph stays as it was. The usual causes are an unclosed bracket, a decimal comma instead of a point (`0,5` instead of `0.5`) or a variable other than x and y.

Common problems

SYMPTOM	WHAT TO DO
The curve is squashed against the axis	The minimum price is too low. Raise it to 1 and the vertical axis comes down with it.
The axis moves every time I touch something	Switch on **Fixed vertical axis**.
The number box does nothing	The range boxes apply when you press `Enter` or leave the box, not while you type.
The recorded points vanish	You changed the swept variable. Each variable keeps its own collection of points.
The box for typing φ does not appear	You must first choose which variable is the linear one, by pressing one of the two options.
It says “φ may only depend on x”	You put a y inside φ. The curved half of a quasilinear function depends on one variable only.

Open the applet: Demand Functions. It all runs in the browser, and once loaded, offline too.

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